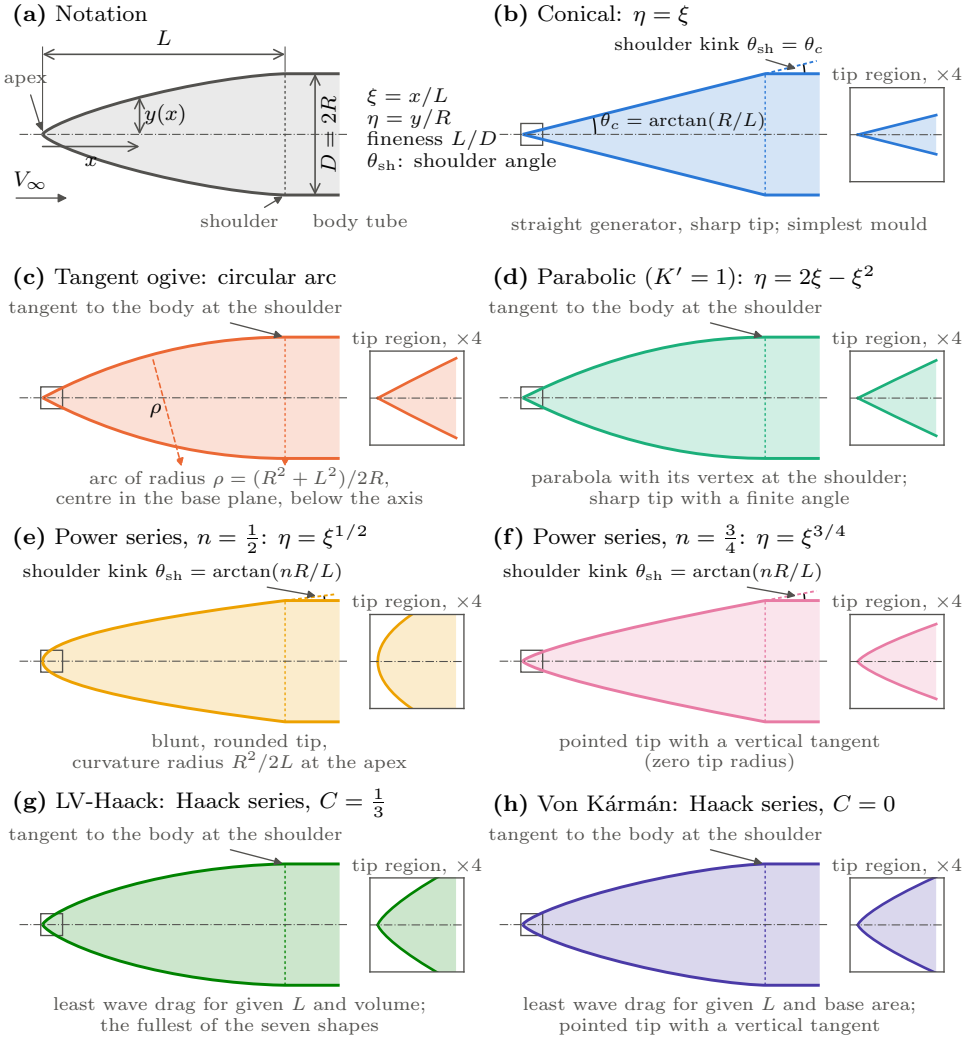

(a) Notation
L
apex
y(x)
x
2R
D = 2R
V∞
shoulder
body tube
ξ = x/L
η = y/R
fineness L/D
θsh: shoulder angle
(b) Conical: η = ξ
shoulder kink θsh = θc
θc = arctan(R/L)
tip region, ×4
straight generator, sharp tip; simplest mould
(c) Tangent ogive: circular arc
tangent to the body at the shoulder
ρ
tip region, ×4
arc of radius ρ = (R² + L²)/2R,
centre in the base plane, below the axis
(d) Parabolic (K′ = 1): η = 2ξ − ξ²
tangent to the body at the shoulder
tip region, ×4
parabola with its vertex at the shoulder;
sharp tip with a finite angle
(e) Power series, n = 1/2: η = ξ^1/2
shoulder kink θsh = arctan(nR/L)
tip region, ×4
blunt, rounded tip,
curvature radius R²/2L at the apex
(f) Power series, n = 3/4: η = ξ^3/4
shoulder kink θsh = arctan(nR/L)
tip region, ×4
pointed tip with a vertical tangent
(zero tip radius)
(g) LV-Haack: Haack series, C = 1/3
tangent to the body at the shoulder
tip region, ×4
least wave drag for given L and volume;
the fullest of the seven shapes
(h) Von Kármán: Haack series, C = 0
tangent to the body at the shoulder
tip region, ×4
least wave drag for given L and base area;
pointed tip with a vertical tangent